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(57) Abstract :
ABSTRACT MACHINE LEARNING BASED PERSONALIZED MENSTRUAL CYCLE PREDICTION AND PCOS RISK ASSESSMENT SYSTEM The present invention discloses a personalized menstrual health management system (100) integrating machine learning algorithms for accurate menstrual cycle prediction and non-invasive PCOS/PCOD risk assessment. The system comprises a data collection module (110) receiving user-specific physiological and lifestyle data, a data preprocessing module (120) performing feature engineering to generate temporal patterns and symptom severity scores, and a machine learning module (130) including a menstrual cycle prediction model (132) implemented using Random Forest Regressor achieving plus or minus one day prediction accuracy, and a PCOS/PCOD risk assessment model (134) implemented using Random Forest Classifier with SMOTE balancing unit (136) and threshold tuning unit (138) achieving 0.75 recall for early detection. A recommendation engine (150) provides personalized lifestyle guidance based on analysis results. The invention addresses limitations of calendar-based tracking methods and provides accessible, data-driven menstrual health monitoring with preventive care capabilities without requiring specialized medical equipment or clinical intervention.

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